

Instructions

There are **25 total points**. When asked to provide your answer within a figure or table, be careful to not exceed box boundaries. Bubbles must be filled out completely: ● is correct, ✓ ● ✗ are incorrect. All answers must be given within the provided circles, answer boxes, figures or tables.

1. [1 point]: Write your full name in the box to acknowledge the instructions.

Supervised Learning

2. [2 points]: Linear regression is suitable for modeling which types of relationships?

- ☐ Polynomial relationships ☐ Any non-linear relationship
☐ First-degree linear relationships ☐ None of the above

3. [2 points]: Logistic regression is best suited for which type of task?

- ☐ Non-linear prediction ☐ Predicting probabilities for distinct outcomes
☐ Clustering ☐ Dimensionality reduction

4. [2 points]: What is one advantage of using a random forest over a single decision tree?

- ☐ Simpler to interpret ☐ Reduced risk of overfitting
☐ Requires less computational power ☐ All of the above

5. [2 points]: Describe **two ways** that random forests introduce randomness into the model over basic decision trees.
(Answer inside the box)

Unsupervised Learning

6. [1 point]: Which dimensionality reduction technique is most suited for visualization? (Select the single best answer.)

- ☐ PCA ☐ t-SNE
☐ Random Forest ☐ Logistic Regression

7. [3 points]: Describe how dimensionality reduction can improve model robustness.

(Answer inside the box)

Initials: ____

8. [1 point]: Which clustering technique is density-based? (Select the single best answer.)

☐ k-means

☐ GMM

☐ DBSCAN

☐ Hierarchical clustering

9. [2 points]: K-means clustering always creates the same clusters when run multiple times.

☐ Yes ☐ No

10. [4 points]: Consider the dendrogram and the cluster options below. Which of the following are valid clusters? (Select all that apply.)

☐ {A, B}

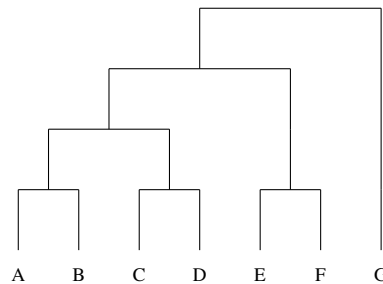
☐ {C, E}

☐ {C, D}

☐ {E, F, G, A}

☐ {E, F, G}

☐ {A, C, G}



Generative AI

11. [3 points]: Describe how a generative model for network traffic traces can improve model robustness for a model like quality of experience (QoE) inference.

(Answer inside the box)

Feedback

12. [1 point]: Interest (1=Boring!; 10=Amazing!):

Difficulty (1=Too easy; 10=Too hard):

13. [1 point]: 1. One thing you like. 2. One topic you would have wanted to see covered that wasn't:

(Answer inside the box)

Initials: _____